Explainable Machine Learning for Predicting Neck Pain and Musculoskeletal Risk from Smartphone Usage

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***Abstract*—This study builds an explainable machine learning workflow that classifies smartphone users as "high risk" or "low risk" of neck pain and musculoskeletal (MSK) strain from usage and lifestyle habits and shows which factors the trained models rely on. The public smartphone-usage dataset (7,500 records, 16 columns) has no recorded neck-pain outcome, so a documented, ergonomics-inspired proxy target (*msk_risk*, 43.1% high risk) is generated by seeded sampling from a logistic risk score, and 3 features are engineered, giving 18 inputs. Random Forest and a gradient-boosted tree model are tuned with randomized search and 5-fold stratified cross-validation (ROC-AUC) and compared on a stratified held-out test set of 1,500 records. Random Forest performed best (accuracy 0.7607, F1-score 0.7228, ROC-AUC 0.8232, PR-AUC 0.7867), only 0.013 below the 0.836 ceiling set by the label noise. Shapley-value explanations show that daily screen time, stress level and sleep hours are the most influential factors. In the reported run, scikit-learn's gradient boosting stood in for XGBoost and a Monte-Carlo Shapley estimator stood in for SHAP; the released code uses XGBoost and SHAP. Because the target is synthetic, the results demonstrate the method and the software pipeline rather than real-world epidemiology, and must not be used for clinical or diagnostic decisions.**

***Keywords*—Neck Pain, Musculoskeletal Disorders, Smartphone Usage, Explainable Machine Learning, SHAP, Random Forest, XGBoost, Proxy Target, Target Leakage.**

I. INTRODUCTION

Smartphones are used for many hours a day for messaging, social media, gaming, study and work. Holding a phone below eye level tilts the head forward, and the load on the cervical spine rises sharply with the angle of flexion [2]. Systematic reviews report associations between long or repetitive mobile touch-screen use and neck, shoulder and thumb symptoms [3], a pattern popularly called "text neck" [1]. Short sleep and high stress are also widely discussed as factors that reduce recovery and increase muscle tension. If risky usage patterns can be recognised early, simple advice such as breaks, posture correction and better sleep can be given before pain becomes chronic.

Machine learning can learn how such usage and lifestyle factors combine from tabular data, but many models behave as black boxes, and a model that looks accurate may rely on a misleading shortcut. This is the motivation for explainable machine learning. SHAP (SHapley Additive exPlanations) [6], built on Shapley values from cooperative game theory [13] and computed exactly and quickly for tree ensembles by TreeSHAP [7], assigns every feature a signed contribution to each individual prediction.

The present study demonstrates a complete, small-scale explainable workflow for predicting neck-pain/MSK risk from smartphone usage. The public smartphone-usage dataset (7,500 records) records smartphone behaviour and addiction level but no neck-pain column. A documented proxy target is therefore created, the identifier columns and a duplicate label are excluded to prevent leakage, and two tree-based classifiers—Random Forest [4] and gradient boosting in the XGBoost family [5]—are tuned and compared. The better model is then interpreted with Shapley values.

An important point must be stated clearly at the outset: the target is a synthetic proxy, not measured neck pain. The scores and feature rankings reported here show how well the algorithms recover a designed relationship; they do not describe how well neck pain could be detected in real people. The main contributions are:

- a leakage-free pipeline with a reproducible, openly documented proxy target and 18 numeric features;
- a tuned comparison of Random Forest and a gradient-boosted model with 5-fold stratified cross-validation and a single untouched test set;
- a threshold-sensitivity analysis and a comparison with the oracle ceiling (ROC-AUC 0.836) of the label-generating probability;
- global and local Shapley explanations, read as model reliance and not causation; and
- a transparent statement of the limitations that prevent the results from being read as real-world findings.

II. LITERATURE REVIEW

A. Smartphone Use and Musculoskeletal Symptoms

Hansraj [2] estimated the stress on the cervical spine for different head-flexion angles and showed that the effective load increases markedly as the head tilts forward. Neupane et al. [1] reviewed "text neck" syndrome, and Toh et al. [3] systematically reviewed the association of mobile touch-screen use with musculoskeletal symptoms and exposures. These works motivate screen time, gaming, social-media use, app opens, notifications, sleep and stress as candidate risk indicators.

B. Machine Learning for Tabular Data

Classical ensembles remain a practical choice for tabular data. Random Forest [4] averages many decision trees grown on bootstrap samples with random feature subsets. XGBoost [5] builds trees sequentially, each correcting earlier errors, under a regularised objective. Both are available in the scikit-learn workflow [8]; data preparation uses pandas [10], hyper-parameters are chosen by random search [12], and figures

are drawn with Matplotlib [11]. For imbalanced or moderately imbalanced labels, precision–recall measures complement ROC analysis [9].

C. Explainable Machine Learning

Explanation methods are grouped into global methods, which describe a model as a whole, and local methods, which explain a single prediction. SHAP [6] unifies several local attribution methods through Shapley values [13], and TreeSHAP [7] computes them for tree ensembles. Averaging the absolute value over many people gives a global ranking, while one person's values explain that prediction. Importance describes what a model uses, not what causes an outcome.

D. Summary of Related Work

TABLE I. SUMMARY OF WORKS ON WHICH THIS STUDY BUILDS

Ref.	Focus	Relevance to this study
[1]	Text neck syndrome review	Neck complaints linked to prolonged phone use
[2]	Cervical spine stress vs. head posture	Load rises with the angle of head flexion
[3]	Touch-screen use and MSK symptoms	Association of repetitive use with symptoms
[4]	Random Forests	Ensemble classifier; built-in importance
[5]	XGBoost	Regularised gradient-boosted trees
[6]	SHAP	Local and global explanations
[7]	TreeSHAP	Exact, fast Shapley values for trees
[8]	scikit-learn	Pipelines, models, metrics, search
[9]	Precision–recall plot	Why PR-AUC is reported with ROC-AUC
[10]	pandas	Data loading and preparation
[11]	Matplotlib	Plotting framework
[12]	Random search	Hyper-parameter optimization
[13]	Shapley value	Game-theoretic basis of SHAP

E. Research Gap

Many prediction-focused studies concentrate on classification accuracy. There is value in a small, fully reproducible workflow that (i) documents its target and the way it was generated, (ii) removes identifier and duplicate-label columns to avoid leakage, (iii) tunes with cross-validation on training data only, (iv) explains the chosen model rather than only scoring it, and (v) states plainly what the results do and do not mean. The workflow can later be re-run on a real neck-pain survey by naming its label column in the configuration file.

III. RESEARCH METHODOLOGY

This section describes the research design, dataset, proxy target, features, models, evaluation and software. The overall data flow is shown in Fig. 1.

A. Research Design

The study follows a quantitative approach in two parts. The first is an exploratory analysis of how the high-risk class varies with usage and lifestyle factors. The second tunes, compares and explains two classifiers. All findings are treated as data-driven associations and model behaviour, not as proof of causation or as a substitute for clinical assessment.

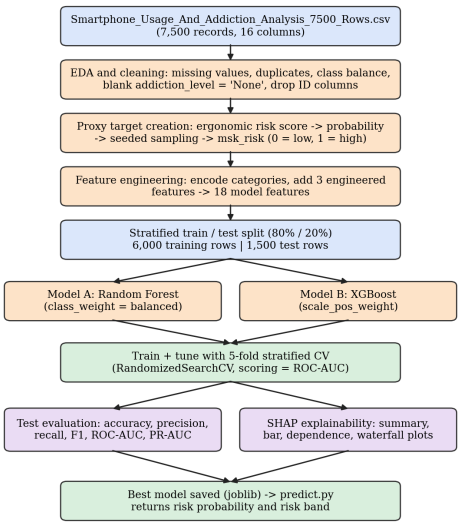


Fig. 1. Data flow of the proposed explainable machine learning pipeline.

B. Dataset Description

The CSV file has 7,500 records and 16 columns with no duplicate rows. Only *addiction_level* has blanks (819 rows), which correspond to users who are not addicted and are filled with "None". The file contains no neck-pain or MSK column. Users are 18–35 years old (mean 26.6) with a mean daily screen time of 7.5 h and a mean sleep of 6.7 h; gender, stress level and the academic/work impact are almost evenly distributed.

TABLE II. GENERAL DESCRIPTION OF THE DATASET

Characteristic	Description
File	Smartphone_Usage_And_Addiction_ Analysis_7500_Rows.csv
Records / columns	7,500 / 16 (no duplicates)
Missing values	addiction_level only (819 rows = None)
Gender	Male 2,553; Other 2,486; Female 2,461
Stress level	High 2,560; Low 2,503; Medium 2,437
Addiction level	Moderate 2,874; Severe 2,434; Mild 1,373; None 819
Model inputs	18 numeric features (15 original + 3 engineered)
Target	msk_risk (synthetic proxy; 1 = high risk)
Nature of target	Generated, educational; not measured pain

TABLE III. DESCRIPTION OF ATTRIBUTES AND THEIR ROLES

Attribute	Role	Description
transaction_id, user_id	Dropped	Unique identifiers
age	Numeric input	Age in years (18–35)
gender	Input (one-hot)	Male, Female, Other
daily_screen_time_hours	Numeric input	Average daily screen time (3–12 h)
weekend_screen_time	Numeric input	Weekend screen time (3.58–14.88 h)
social_media / gaming / work_study_hours	Numeric input	Daily hours per activity
sleep_hours	Numeric input	Sleep per night (4.5–9 h)
notifications / app_opens_per_day	Numeric input	Daily counts
stress_level	Ordinal input	Low 0, Medium 1, High 2
academic_work_impact	Binary input	Phone use affects study/work
addiction_level	Ordinal input	None 0 – Severe 3
addicted_label	Dropped	Duplicates addiction_level
msk_risk	Target (binary)	Proxy label, 43.1% high risk

C. Proxy Target Construction

Because no neck-pain outcome exists, *msk_risk* is generated as shown in Fig. 2. Eight usage columns are standardised

(z-scores) and combined with ergonomic weights; the stress level adds a fixed weight:

$$\text{score} = 0.35\, z_{\text{daily}} + 0.15\, z_{\text{weekend}} + 0.15\, z_{\text{gaming}} + 0.10\, z_{\text{social}} + 0.10\, z_{\text{opens}} + 0.05\, z_{\text{notif}} - 0.20\, z_{\text{sleep}} + 0.05\, z_{\text{age}} + w_{\text{stress}},$$

with w_{stress} = 0 (Low), 0.35 (Medium) and 0.70 (High). The score is converted to a probability $p = 1/(1 + e^{-2.5(\text{score} - c)})$, where c is the 60th percentile of the score, and the label is drawn as 1 if a seeded uniform random number (seed 42) is below p. Sampling makes the label noisy rather than a deterministic formula. The result is 3,232 high-risk (43.1%) and 4,268 low-risk records. Even the true probability p reaches a ROC-AUC of only 0.836 against the sampled label, which is the ceiling for any model.

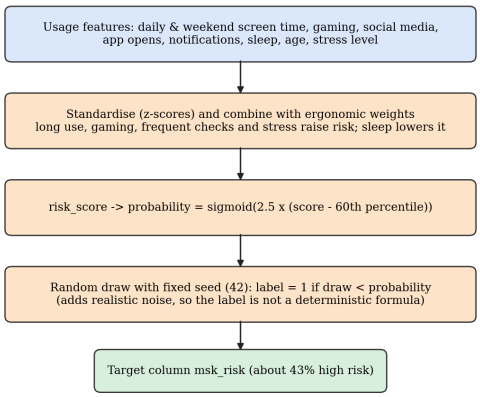


Fig. 2. Creation of the proxy target (*msk_risk*).

D. Feature Engineering and Leakage Prevention

Stress level (0–2) and addiction level (0–3) are ordinal-encoded, academic impact is mapped to 0/1 and gender is one-hot encoded. Three engineered features are added: *non_work_screen_hours* (social + gaming), *weekend_screen_increase* (weekend − daily) and *checks_per_screen_hour* (app opens / max(daily, 0.5)), giving 18 numeric features; trees need no scaling. The identifier columns and *addicted_label* are dropped so that they cannot leak information. The data are split once, before any tuning, into 6,000 training and 1,500 test records (80% / 20%, stratified, seed 42); the high-risk share is 43.1% in training (2,586 cases) and 43.1% in testing (646 cases).

E. Machine Learning Approach

Two classifiers are compared. Random Forest [4] averages the probabilities of many trees grown on bootstrap samples with random feature subsets and uses class_weight = “balanced”. The boosted model builds small trees one after another, each fitted to the errors (gradient and curvature of the log-loss) of the earlier ones, minimising a regularised objective $\text{Obj} = \Sigma\, \text{loss}(y, \hat{y}) + \Sigma\, (\gamma T + \tfrac{1}{2}\lambda \|w\|^2)$ [5]; the released code uses XGBoost with scale_pos_weight = 1.32 (low-risk count / high-risk count). In the run reported here, XGBoost could not be installed, so scikit-learn’s histogram-based gradient boosting [8], a model of the same family, is used and is called “Gradient Boosting (sklearn)”. A record is predicted high risk when $p \geq 0.50$.

Hyper-parameters are tuned with RandomizedSearchCV [12], 15 candidates per model and 5-fold stratified cross-validation (shuffle, seed 42) on the training data only, scored by ROC-AUC. The search spaces are in Table IV.

TABLE IV. HYPER-PARAMETER SEARCH SPACES

Model	Search space
Random Forest	n_estimators 200–600; max_depth 4–19; min_samples_leaf 1–14; max_features {sqrt, log2, 0.5}; class_weight = balanced
Gradient Boosting (sklearn)	max_iter 150–600; max_depth 2–7; learning_rate 0.01–0.21; l2_regularization 0–5; min_samples_leaf 10–79; max_features 0.6–1.0; class_weight = balanced

F. Evaluation and Explainability

Model quality is measured at the 0.50 threshold with accuracy, precision, recall and F1-score, and over all thresholds with ROC-AUC and PR-AUC [9], where TP, TN, FP and FN denote true and false positives and negatives: precision = TP/(TP + FP), recall = TP/(TP + FN) and F1 = 2PR/(P + R). Models are ranked by test ROC-AUC, with F1 as the tie-breaker, and each model is evaluated once on the test set. The best model is explained with Shapley values on a sample of 300 test records: the summary (beeswarm) and bar plots give the global picture, a dependence plot shows the effect of the most important feature, and waterfall plots explain single predictions. Shapley values add up to the prediction, $f(x) = \varphi_0 + \Sigma\, \varphi_i$ [6], [7].

G. Software Implementation

The workflow is implemented in Python 3 with pandas [10], NumPy, scikit-learn [8], XGBoost [5], SHAP [6], Matplotlib [11], seaborn and joblib. The code is split into *config.py*, *make_target.py*, *features.py*, *eda.py*, *train.py*, *explain.py* and *predict.py*; the label column of a real survey can be named in *config.py* to skip the proxy target. It runs on an ordinary laptop (64-bit CPU, 4 GB RAM minimum).

IV. EXPLORATORY ANALYSIS OF THE DATASET

A. Class Distribution and Usage Differences

The classes are only mildly imbalanced (about 1.3 low-risk users for every high-risk user), so accuracy is meaningful, and precision, recall, F1, ROC-AUC and PR-AUC are reported as well. High-risk users average 8.8 h of daily screen time against 6.5 h for low-risk users, sleep 6.5 versus 6.9 h and open apps 103 versus 94 times a day. Table V shows that the high-risk rate rises steeply with screen time (12.6% for 4 h or below to 73.2% above 10 h) and with stress (28.1% for low to 59.0% for high stress), and falls with more sleep (53.5% for 5 h or below to 32.9% above 8 h). Gender shows almost no difference (43.8% male, 42.4% female). These differences follow from how the proxy label was designed.

TABLE V. HIGH-RISK RATE BY FACTOR

Factor	Group	Records	High-risk rate
Daily screen time (h)	4 or below	868	12.56%
	4–6	1,672	20.63%
	6–8	1,616	37.56%
	8–10	1,629	56.23%
	Above 10	1,715	73.18%
Sleep (h)	5 or below	800	53.50%
	5–6	1,696	51.30%
	6–7	1,748	44.16%
	7–8	1,661	38.41%
	Above 8	1,595	32.85%
Stress level	Low / Medium / High	2,503 / 2,437 / 2,560	28.13 / 41.77 / 58.98%
Addiction level	None / Mild	819 / 1,373	17.22 / 23.02%
	Moderate / Severe	2,874 / 2,434	50.35 / 54.56%
Gender	Female / Male / Other	2,461 / 2,553 / 2,486	42.38 / 43.79 / 43.08%
Academic/work impact	No / Yes	3,753 / 3,747	42.42 / 43.77%

B. Correlations

The strongest linear correlations with the target are *daily_screen_time_hours* (+0.449), *weekend_screen_time* (+0.438) and *addiction_level* (+0.272); sleep is negatively correlated (−0.150). The two screen-time variables are correlated with each other (0.96), which is not a problem for tree-based models, and addiction level partly mirrors screen time (0.54). These are associations in the data, not direct causes.

V. MODEL EVALUATION RESULTS

A. Hyper-parameter Tuning

TABLE VI. BEST CROSS-VALIDATED ROC-AUC AND HYPER-PARAMETERS

Model	CV ROC-AUC	Best hyper-parameters
Random Forest	0.8240	max_depth = 10, max_features = sqrt, min_samples_leaf = 9, n_estimators = 366
Gradient Boosting (sklearn)	0.8228	l2_regularization = 3.9509, learning_rate = 0.1312, max_depth = 3, max_features = 0.7434, max_iter = 190, min_samples_leaf = 37

Random Forest scored slightly higher in cross-validation (0.8240 versus 0.8228); the two models are close.

B. Model Comparison

Table VII and Fig. 3 show the held-out test performance on 1,500 records (646 high risk) at the 0.50 threshold. Random Forest is the best model: it has the higher ROC-AUC (0.8232 versus 0.8210) and an F1-score of 0.7228. The ROC-AUC difference of 0.0022 is small and, with a single split, within the noise; the approximate 95% intervals for the accuracies (0.76 ± 0.022 and 0.75 ± 0.022) overlap almost completely. The true generating probability reaches ROC-AUC 0.836, so the best model is only 0.013 below this ceiling: both models recover almost all of the learnable signal and further tuning cannot help much.

TABLE VII. TEST-SET EVALUATION, 1,500 RECORDS (THRESHOLD 0.50; ROC = ROC-AUC, PR = PR-AUC)

Model	Acc.	Prec.	Rec.	F1	ROC	PR
Random Forest	0.7607	0.7211	0.7245	0.7228	0.8232	0.7867
Gradient Boosting (sklearn)	0.7520	0.7039	0.7322	0.7178	0.8210	0.7848

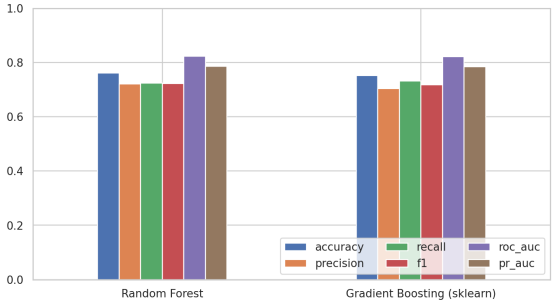


Fig. 3. Test-set metrics of the two models.

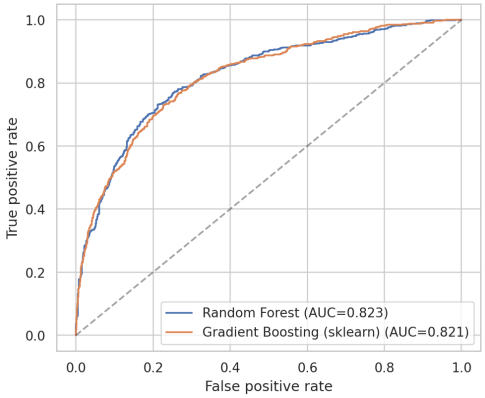


Fig. 4. ROC curves of the two models.

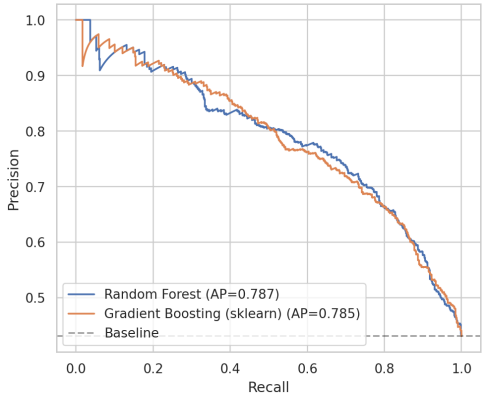


Fig. 5. Precision–recall curves of the two models.

A ROC-AUC of about 0.82 means that, for a random high-risk user and a random low-risk user, the model ranks the high-risk user higher about 82% of the time. The PR-AUC of 0.79 is about 1.8 times the baseline of 0.431 (the high-risk share), which shows clear but not perfect predictive power.

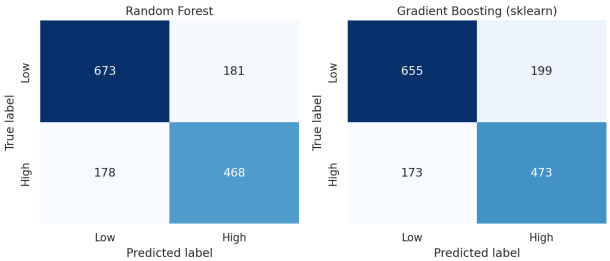


Fig. 6. Confusion matrices of the two models on the test set.

Fig. 6 shows that Random Forest produced 673 true negatives, 181 false positives, 178 false negatives and 468 true positives; the boosted model produced 655, 199, 173 and 473. Table VIII gives the class-wise report of Random Forest, with similar quality in both classes.

TABLE VIII. CLASS-WISE REPORT OF THE BEST MODEL (RANDOM FOREST)

Class	Precision	Recall	F1-score	Support
Low risk	0.79	0.79	0.79	854
High risk	0.72	0.72	0.72	646
Accuracy	–	–	0.76	1,500

C. Effect of the Decision Threshold

Table IX shows how the metrics change when the probability threshold is moved. Lowering it from 0.7 to 0.3 raises recall from 0.43 to 0.89 while precision falls from 0.83 to 0.58; missed high-risk users fall from 367 to 70 but false positives rise from 58 to 411. A health-awareness screening use would favour a lower threshold so that few people are missed, whereas a use that must limit unnecessary follow-up would favour a higher one. The default 0.50 is used for the main results.

TABLE IX. THRESHOLD SENSITIVITY OF RANDOM FOREST (TEST SET)

Thr.	Acc.	Prec.	Recall	F1	FP	FN
0.3	0.6793	0.5836	0.8916	0.7055	411	70
0.4	0.7327	0.6485	0.8282	0.7274	290	111
0.5	0.7607	0.7211	0.7245	0.7228	181	178
0.6	0.7527	0.7745	0.6006	0.6765	113	258
0.7	0.7167	0.8279	0.4319	0.5677	58	367

D. Built-in Feature Importance

Random Forest ranks *daily_screen_time_hours*, *weekend_screen_time* and *stress_level* highest (Fig. 7), while the boosted model (permutation importance, drop in ROC-AUC) ranks *daily_screen_time_hours*, *stress_level* and *sleep_hours* highest (Fig. 8). The ranking agrees with Table V, but built-in importance shows only what a model relies on, not the direction of the effect, which is why Shapley values are used next.

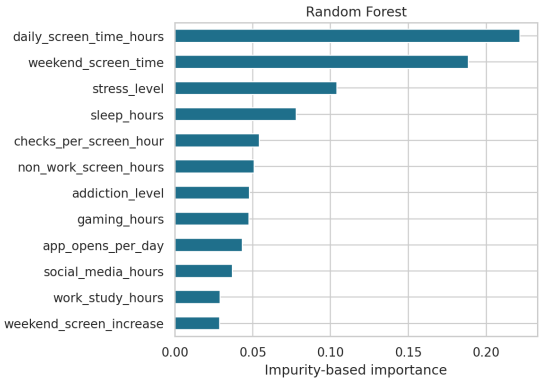


Fig. 7. Random Forest impurity-based feature importance.

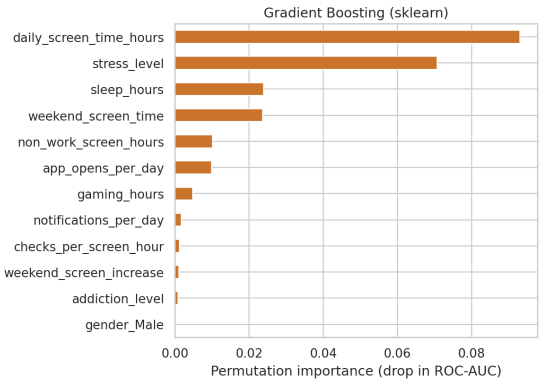


Fig. 8. Gradient Boosting (sklearn) permutation importance.

E. Explainability with Shapley Values

Each dot in Fig. 9 is one test person; the colour shows the feature value (red = high, blue = low) and the horizontal position shows how much that feature pushed the prediction up (right) or down (left). Daily screen time, stress level and sleep hours have the widest spread. High values of screen-time and usage variables lie on the right (more risk), while high sleep hours lie on the left (less risk). This agrees with the ergonomic logic used to build the label, which shows that the model has learned the intended relationships. The Shapley figures in this paper were computed for the boosted model with a Monte-Carlo permutation estimator of the quantity that SHAP computes; running the released code with the *shap* package regenerates them with TreeSHAP [7] for either model.

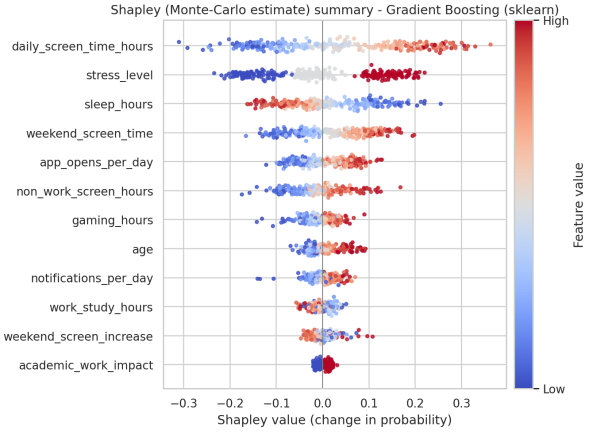


Fig. 9. Shapley-value summary (beeswarm) plot of Gradient Boosting (sklearn).

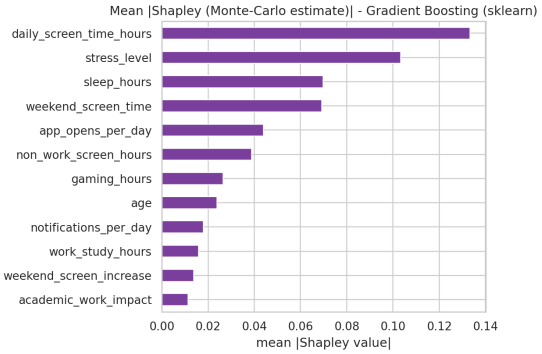


Fig. 10. Mean |Shapley value| per feature.

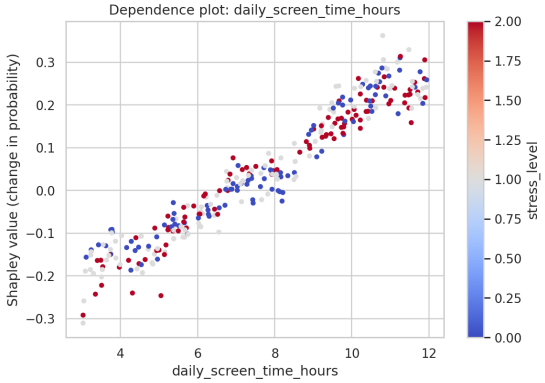


Fig. 11. Dependence plot of *daily_screen_time_hours*; the colour shows stress level.

TABLE X. TOP FEATURES BY MEAN |SHAPLEY VALUE|

Rank	Feature	Mean |Shapley value|
1	daily_screen_time_hours	0.1334
2	stress_level	0.1034
3	sleep_hours	0.0698
4	weekend_screen_time	0.0692
5	app_opens_per_day	0.0439
6	non_work_screen_hours	0.0388
7	gaming_hours	0.0266
8	age	0.0238

The waterfall plots in Figs. 12 and 13 start at the average model output and add the contribution of each feature to reach the person's own score. For the highest-risk person several usage features and stress push the score up; for the lowest-risk person low usage and long sleep push it down. This is the local explanation that a single feature-importance bar chart cannot give.

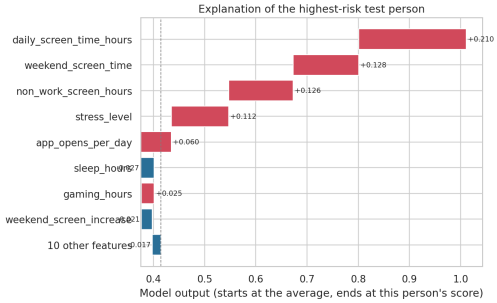


Fig. 12. Waterfall explanation of the highest-risk test person.

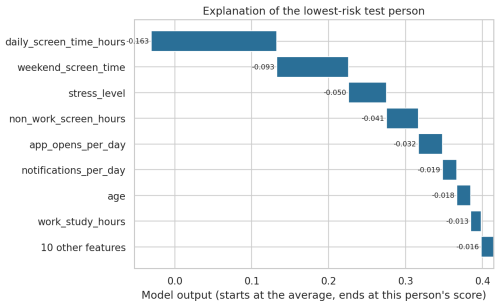


Fig. 13. Waterfall explanation of the lowest-risk test person.

F. Predictions for Example Users

TABLE XI. PREDICTIONS FOR THREE EXAMPLE USERS

User	Key inputs	P(GB)	P(RF)	Risk	Top drivers
Heavy user	10.5 h, sleep 5.0 h, High stress, Severe	0.982	0.966	HIGH	daily screen time (+0.16); stress (+0.09); weekend screen (+0.08)
Light user	3.0 h, sleep 8.0 h, Low stress, None	0.006	0.020	LOW	daily screen time (−0.12); non-work screen (−0.08); stress (−0.05)
Average user	7.5 h, sleep 6.7 h, Medium, Moderate	0.469	0.449	LOW	age (−0.05); daily screen time (+0.04); work/study (+0.03)

TABLE XII. SUMMARY OF MAJOR ANALYTICAL RESULTS

Analysis	Main finding
Proxy target	3,232 high risk (43.1%); oracle ROC-AUC 0.836
Best CV ROC-AUC	Random Forest 0.8240 (GB 0.8228)
Best model (test)	Random Forest
Accuracy / F1	0.7607 / 0.7228
ROC-AUC / PR-AUC	0.8232 / 0.7867
Gap to oracle ceiling	0.013 ROC-AUC
Top Shapley features	Daily screen time, stress, sleep

VI. DISCUSSION AND HUMAN INTERPRETATION

A. What the Models Rely On

Both models rely mainly on daily screen time, stress level and sleep hours, with weekend screen time, app opens, non-work screen time, gaming and age contributing less. This is consistent with the ergonomic literature [1]–[3], but it is also exactly what the proxy label was designed to contain, so it is evidence that the pipeline recovers the designed relationship, not evidence about real neck pain. Gender and academic/work impact are weak indicators.

B. Choice Between the Two Models

Random Forest was selected because it had the higher test ROC-AUC. However, the gap of 0.0022 is small, only one split of 1,500 records is used, and the models differ by less than 0.01 ROC-AUC in cross-validation as well. The choice is a ranking and not strong proof that Random Forest is better; repeated cross-validation would give more stable estimates.

C. Reading Explanations and Probabilities Responsibly

Shapley values describe reliance, not causation, and correlated features such as addiction level and screen time share contributions. The probabilities come from a synthetic label and are not calibrated real-world risks; Platt scaling or isotonic regression would be needed for dependable percentages. The output is a screening-level risk ranking, not a diagnosis.

VII. LIMITATIONS AND FUTURE SCOPE

A. Limitations

1. **Synthetic proxy target:** the dataset has no MSK outcome, so test metrics measure recovery of a designed relationship (the generating formula itself reaches ROC-AUC 0.836) and say nothing about real neck pain.
2. **Synthetic-looking dataset:** for 60.7% of users the social-media, gaming and work/study hours add up to more than the daily screen time, which is not physically consistent.
3. **Stand-in tools in the reported run:** scikit-learn gradient boosting replaced XGBoost and a Monte-Carlo Shapley estimator replaced SHAP; numbers for the boosted model and the Shapley figures may differ slightly with the real packages.
4. **Single split:** one train–test split of 1,500 records; the two models differ by less than 0.01 ROC-AUC.
5. **Label noise:** precision and recall are moderate even in this ideal setting; a real-world label would be noisier still.
6. **Missing factors:** posture, neck angle, phone-holding style, device size, exercise, occupation and previous injuries are not in the data.
7. **Explanation scope:** Shapley values show reliance, not causation, and overlap for correlated features.

B. Future Scope

Future work includes collecting a real neck-pain/MSK questionnaire (for example the Neck Disability Index) together with objective usage logs so that the proxy label can be replaced by measured outcomes, and adding posture and device-angle features from phone sensors. Probability calibration, repeated cross-validation and external validation on another population are required. Finally, the model and its SHAP explanations can be packaged into a web or mobile app that gives users personalised advice on breaks, posture and sleep.

VIII. CONCLUSION

This study implemented an end-to-end, leakage-free explainable machine learning workflow for predicting neck-pain/MSK risk from smartphone usage on 7,500 records.

Because the dataset has no recorded outcome, a documented ergonomics-inspired proxy target (43.1% high risk) was created. Random Forest and a gradient-boosted model were tuned with 5-fold stratified cross-validation and evaluated once on an untouched 1,500-record test set. Random Forest performed best (ROC-AUC 0.8232, PR-AUC 0.7867, F1-score 0.7228, accuracy 0.7607), very close to the 0.836 ceiling set by the label noise, while the two models differed by only 0.0022 ROC-AUC. Shapley analysis showed that daily screen time, stress level and sleep hours are the most influential factors, in line with the ergonomic assumptions behind the label.

The workflow demonstrates the full pipeline and its explanations, but since it is trained on a proxy label, its predictions must be treated as demonstration risk rankings rather than diagnostic verdicts. It should be re-run on real survey data before any substantive conclusion is drawn.

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